

Seneca College

School of Information Technology Administration & Security - SITAS

CSN405 – Wireless Networks

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Lab 7: Basic Configuration of Controller WLC- Aruba7005

Objective:

The main objective of this lab is to familiarize yourself with the configuration of Wireless Controller Aruba 7005.

Required Material:

- 1) 1 x Aruba 7005 Controller
- 2) 1 x Switch Aruba 2530 (to provide PoE power connection to the Controller)
- 3) PC
- 4) 4 x Straight Through Ethernet cables. One of them is used to provide power (PoE) to the Controller.
- 5) 1 x console cable

Lab Submission Guidelines:

1. This is a group work, **but** each student must submit his/her report separately on Bb.
2. Submissions by email will not be considered. Zero mark will be granted for submissions after due date.
3. In order to submit your report, edit this document and answer the questions by using **bold dark blue** fonts. Take sufficient number of screenshots for different steps and insert them in the proper locations of the lab document.
4. After finishing your lab, submit your file by writing your name on the file as follows: **Your First Name. Your Last Name - Lab 7**. Drop your file in the corresponding drop box.

Introduction:

The Aruba 7005 Controller is a wireless LAN controller that connects, controls, and intelligently integrates wireless Access Points (APs) and Air Monitors (AMs) into a wired LAN system. The 7005 controller includes four Ethernet ports, one Console port, one Mini-USB console port, and one USB port. This device supports up to 16 APs and 1024 users.

The 7005 controller requires ArubaOS 6.4.1.0 or later. The controller AoS (Aruba operating System) that is used in this controller is 8.0.3.9

Controller 7005 Components:

1. In the Aruba 7005 controller, the orange numbering on port 0 indicates that it is a PoE powered device (PoE-PD) port and the gray numbering on ports 1, 2, and 3 indicate that

they are non-PoE ports. So, provide power to the controller by connecting a straight through cable from any port on the switch to PoE port (orange colored) number 0 ONLY.

2. **Figure 1** shows the front panel of the Aruba 7005 controller and **Figure 2** shows the back panel of the Aruba 7005 controller.

Figure 1 Front Panel of the 7005 Controller

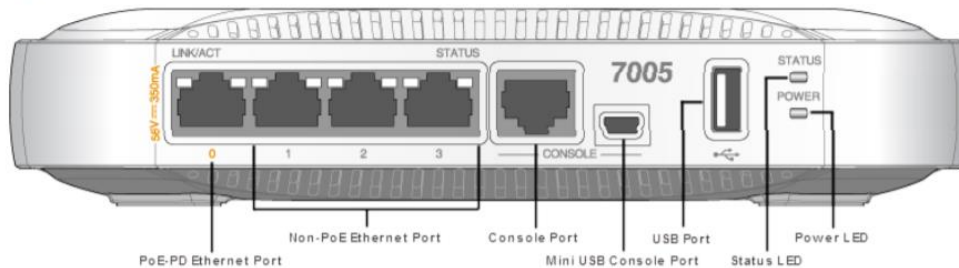
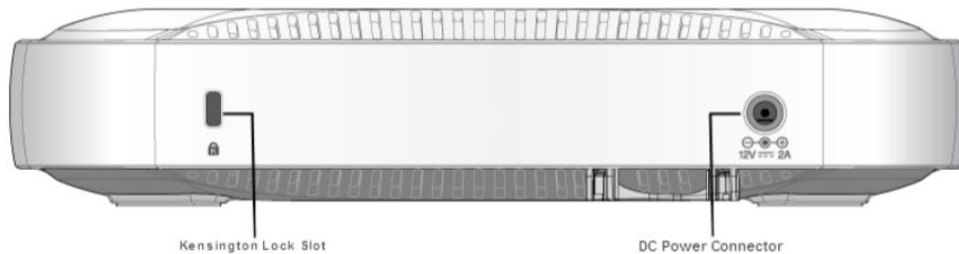


Figure 2 Back Panel of the 7005 Controller



Ethernet Port LEDs

The Aruba 7005 controller is equipped with four 10/100/ 1000BASE-T Gigabit Ethernet ports (0 to 3). Each 10/100/1000BASE-T Ethernet port is equipped with two LEDs that allow basic monitoring of link/port status and activity.

- **LINK/ACT:** Placed on the left side of the port, this LED displays the link status and activity of the port.
- **STATUS:** Placed on the right side of the port, this LED displays the status of the port based on the CLI.

The following table describes the LED behavior for each mode:

Table 1 10/100/1000BASE-T Ethernet Port LEDs

LED	Function	Mode	Indicator	Status
LINK/ACT	Link status	N/A	Green (Solid)	Link has been established
			Green (Blinking)	Port is transmitting or receiving data
			Off	No link on port
STATUS	Port status	Speed	Green (Solid)	1000 Mbps
			Off	10/100 Mbps

Console Port

The serial console port allows connecting the controller to a serial terminal or a laptop for direct local management. This port is an RJ-45 female connector with the pinouts described in **Figure 3**. Connect it directly to a terminal or terminal server (PuTTY) using an Ethernet cable.

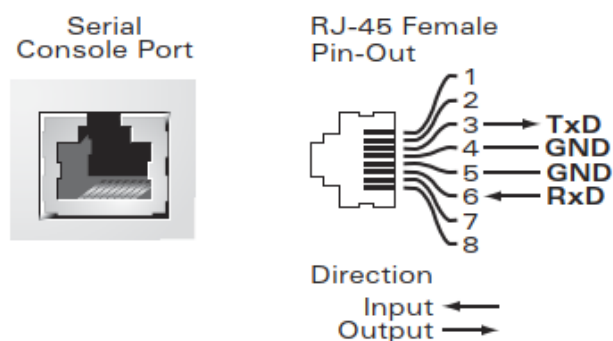


Figure 3: Serial Console Port Pin-Out

The communication settings for the Console port is shown in the following table:

Table 2 Console Terminal Settings

Baud Rate	Data Bits	Parity	Stop Bits	Flow Control
9600	8	None	1	None



Do not connect the Console port to an Ethernet switch or a PoE power source. This may damage the controller.

Power and Status LEDs

The front panel also includes power and status LEDs that provide basic monitoring of the overall status of the Aruba 7005 controller.

Table 3 Power and Status LEDs

LED	Function	Indicator	Status
Power	System powers	Green (Solid)	Powered from DC adapter
		Amber (Solid)	Powered from PoE source
		Off	Power Off
Status	System status	Green (Solid)	Operational
		Green (Blinking)	Device is loading software
		Amber (Blinking)	Major Alarm
		Amber (Solid)	Critical Alarm
		Off	No power

Network Topology:

Build up the following network:



Figure 4: WLC-7005 Controller with Switch Aruba 2530

The straight through cable is used to provide power to the controller and the console cable to configure the device.

Note: You need to add also a straight through cable (not shown in the figure) between the wireless controller (Ethernet port, #1 for example) to the NIC card on PC for data communication. This means you will have two connections, one for the console and one for the data.

Configuration Steps:

In this lab, we will configure 7005 controller by using two methods:

Part 1: Configuration using CLI

1. Access the controller from the PC using PuTTY.
2. Once you are done with the connections, you must get the following window after the controller is booted up:

```
Enter Option (partial string is acceptable): Using port gigabitethernet 0/0/3 for auto provisioning
Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning...
'enable-debug'      : Enable auto-provisioning debug logs
'disable-debug'     : Disable auto-provisioning debug logs
'mini-setup'        : Start mini setup dialog. Provides minimal customization and requires DHCP server
'full-setup'        : Start full setup dialog. Provides full customization
'static-activate'   : Provides customization for static or PPPOE ip assignment. Uses activate for master information
Enter Option (partial string is acceptable):
```

3. If the controller is not on default factory setting, then type the following command:
#write erase

Note: User Name is **admin** and Password is **admin123**

```
User: admin
Password:
(POD-H) [mynode] #write erase
All the configuration will be deleted and the controller will be reloaded. Press 'y' to proceed : [y/n]:
```

```
User: admin
Password:
(POD-H) [mynode] #write erase
All the configuration will be deleted and the controller will be reloaded. Press 'y' to proceed : [y/n]: y
uWrite Erase successful

System will now restart!
```

4. Since we configure this controller as a standalone (**Local Mode**), we don't need any **Mobility Master** (MM) controller. Therefore, we choose a **full** manual setup configuration. Such configuration is useful to implement a private wireless small business network.

```
Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning...
'enable-debug'      : Enable auto-provisioning debug logs
'disable-debug'     : Disable auto-provisioning debug logs
'mini-setup'        : Start mini setup dialog. Provides minimal customization and requires DHCP server
'full-setup'        : Start full setup dialog. Provides full customization
'static-activate'   : Provides customization for static or PPPOE ip assignment. Uses activate for master information

Enter Option (partial string is acceptable): full

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): y
```

Note: For the following step, your *group-number* is your POD Letter.

```

Enter System name [Aruba7005]: CSN405-group-number
Enter Switch Role (standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 1
Enter Controller VLAN port [GE 0/0/0]:
Enter Controller VLAN port mode (access|trunk) [access]: access
Enter VLAN interface IP address [172.16.0.254]: 172.16.0.1
Enter VLAN interface subnet mask [255.255.255.0]:
Enter IP Default gateway [none]:
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: n
Enter Country code (ISO-3166), <ctrl-I> for supported list: ca
You have chosen Country code CA for Canada (yes|no)? : y
Enter the controller's IANA Time zone [America/Los_Angeles]:
Enter Time in UTC [17:42:16]:
Enter Date (MM/DD/YYYY) [8/18/2021]:
Enter Password for admin login (up to 32 chars): *****
Re-type Password for admin login: *****

Current choices are:

System name: CSN405-group-number
Switch Role: standalone
Controller VLAN id: 1
Controller VLAN port: GE 0/0/0
Controller VLAN port mode: access
VLAN interface IP address: 172.16.0.1
VLAN interface subnet mask: 255.255.255.0
Option to configure VLAN interface IPV6 address: no
Country code: ca
IANA Time Zone: America/Los_Angeles

If you accept the changes the switch will restart!
Type <ctrl-P> to go back and change answer for any question
Do you wish to accept the changes (yes|no)

```

5. Answer Yes in this step,

```

If you accept the changes the switch will restart!
Type <ctrl-P> to go back and change answer for any question
Do you wish to accept the changes (yes|no)y
INFO: Backing up existing config dir.
Creating configuration... Done.

System will now restart!

```

6. Note that during the process of restarting the Status green LED on the front panel of the controller starts blinking. The following window is what you get once the restarting process is finished:

```

<<<<<   Welcome to Aruba Networks - Aruba A7005   >>>>>

[17:45:59]:Probing for EEPROM devices                [ OK ]
[17:45:59]:Probing for real-time clock                [ OK ]
[17:45:59]:Uncompressing core image files            [ OK ]
[17:47:07]:Extracting corefs                          [ OK ]

[17:47:10]:Enabling watchdog                          [ OK ]
[17:47:10]:Starting device manager

Performing eUSB Flash fast test...                    [ DONE ]
[ OK ]
[17:47:12]:Mounting flash                            [ OK ]
[17:47:12]:Initializing 511MB as swap on zRam0        [ OK ]
[17:47:14]:Turning swap ON on zRAM0                  [ OK ]
[17:47:14]:Checking system inventory                 [ OK ]
[17:47:14]:Installing ancillary FS                   [ OK ]
Performing integrity check on ancillary partition 0   [ OK ]
Extracting Webui files../

[17:47:50]:Reboot Cause: User reboot (Intent:cause:register 78:86:0:2c)
[17:47:50]:Starting syslog service                   [ OK ]
[17:47:51]:Restoring the database                     [ OK ]
[17:47:57]:Generating SSH keys                       [ OK ]
[17:47:57]:Initializing TPM and certificates          [ OK ]
[17:48:26]:Checking for configuration upgrade         [ OK: Version Upgrade]
[17:48:31]:rcS Done(159 sec)

[17:48:31]:Starting OS services                      [ OK ]

[17:48:36]:Initializing GSM                          [ DONE ]
[17:48:38]:Initializing CCM                          [ DONE ]
[17:49:46]:Initializing FPAPPs                       [ DONE ]
[17:50:09]:Initializing CFGM                         [ DONE ]
[17:50:09]:Waiting for Controller IP                  [ DONE ]
[17:50:11]:Initializing AAA                          [ DONE ]
[17:50:11]:Initializing Controller management         [ DONE ]

User: █
```

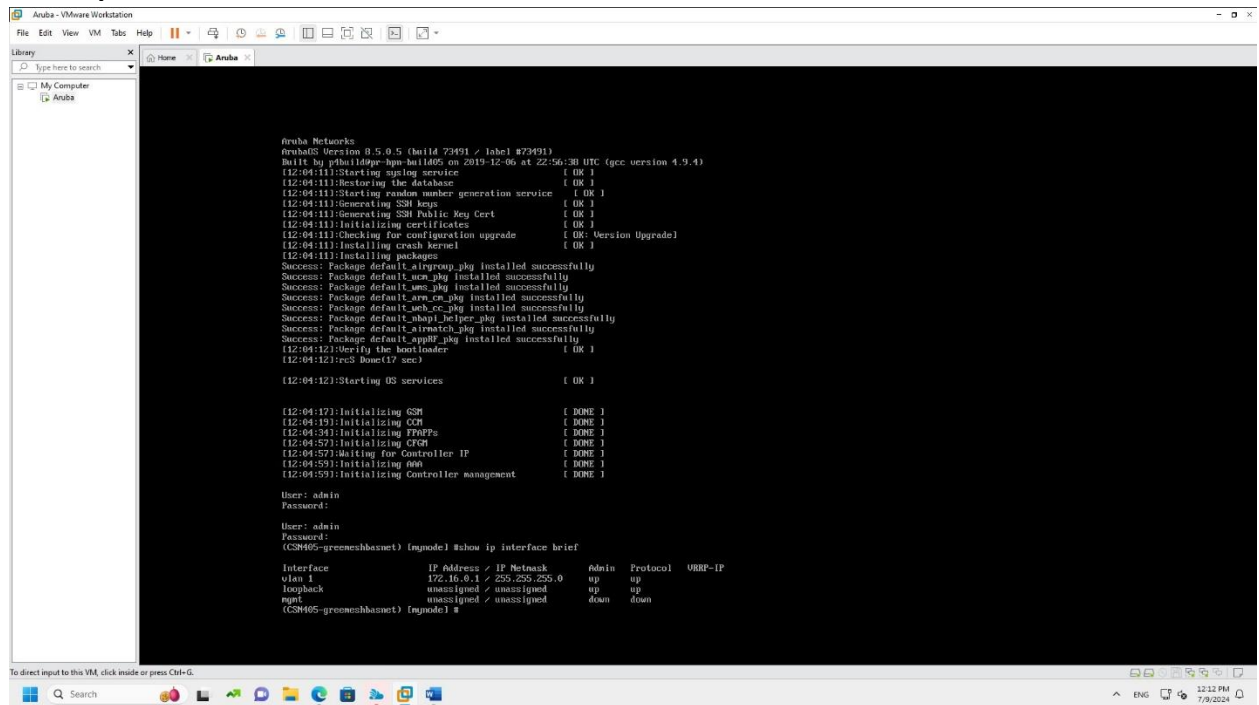
7. Now, enter User Name as **admin** and Password as **admin123**

```

User: admin
Password:
(CSN405-group-number) [mynode] #█
```

8. To demonstrate finishing this lab, issue the following four commands and provide screenshots for each one of them:

1. Show ip interface brief



```
Aruba Networks
ArubaOS Version 8.5.0.5 (build 73491 / label #73491)
Built by yfbuild@hp-hpa-build05 on 2019-12-06 at 22:56:30 UTC (gcc version 4.9.4)
(12-04:11):Starting syslog service [ OK ]
(12-04:11):Restoring the database [ OK ]
(12-04:11):Starting random number generation service [ OK ]
(12-04:11):Generating SSH keys [ OK ]
(12-04:11):Generating SSH Public Key Cert [ OK ]
(12-04:11):Initializing certificates [ OK ]
(12-04:11):Checking for configuration upgrade [ OK: Version Upgrade]
(12-04:11):Installing crash kernel [ OK ]
(12-04:11):Installing packages
Success: Package default_alngrp_pkg installed successfully
Success: Package default_uca_pkg installed successfully
Success: Package default_uaa_pkg installed successfully
Success: Package default_ara_cn_pkg installed successfully
Success: Package default_web_cc_pkg installed successfully
Success: Package default_ahapi_helper_pkg installed successfully
Success: Package default_alnmatch_pkg installed successfully
Success: Package default_appRF_pkg installed successfully
(12-04:12):Verify the bootloader [ OK ]
(12-04:12):rcS Done(17 sec)

(12-04:12):Starting OS services [ OK ]

(12-04:17):Initializing GSM [ DONE ]
(12-04:19):Initializing CCM [ DONE ]
(12-04:34):Initializing FPMPPs [ DONE ]
(12-04:52):Initializing CQM [ DONE ]
(12-04:57):Waiting for Controller IP [ DONE ]
(12-04:59):Initializing AAA [ DONE ]
(12-04:59):Initializing Controller management [ DONE ]

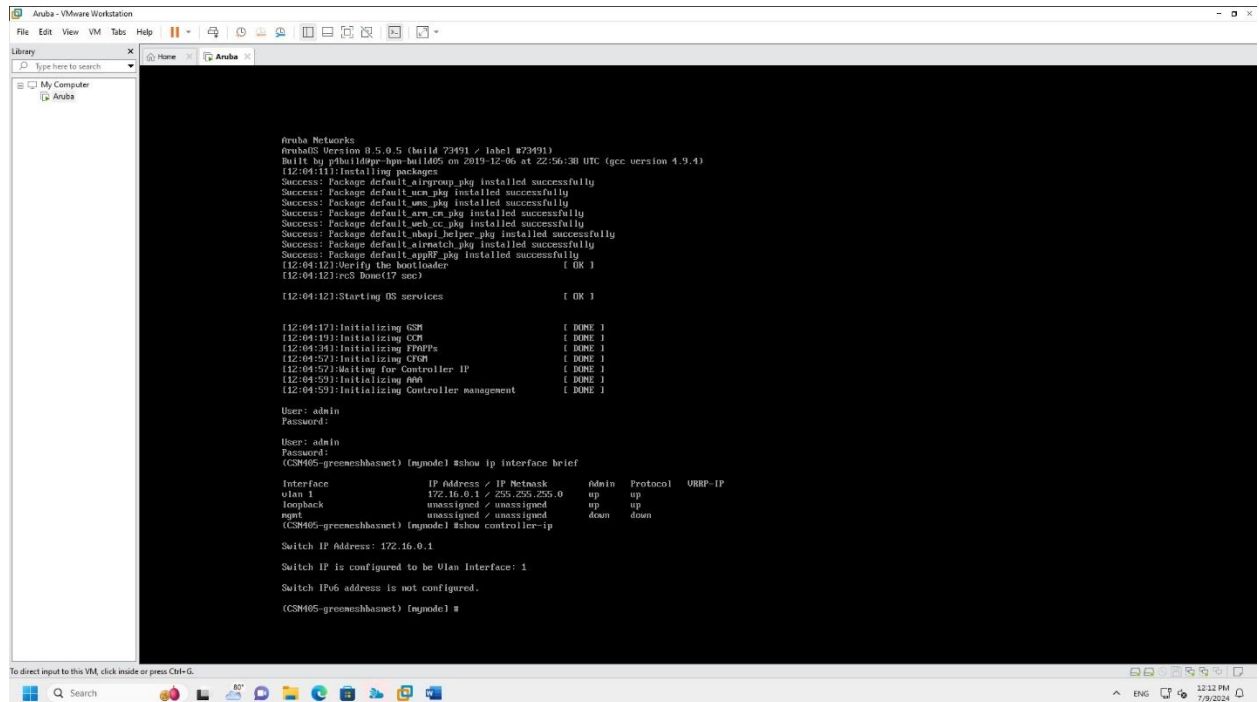
User: admin
Password:

User: admin
Password:

(CSM405-greeneshbasnet) [mgnode] show ip interface brief

Interface IP Address / IP Netmask Admin Protocol VRRP-IP
vlan 1 172.16.0.1 / 255.255.255.0 up up
loopback unassigned / unassigned up up
mgmt unassigned / unassigned down down
(CSM405-greeneshbasnet) [mgnode] #
```

2. Show controller-ip



```
Aruba Networks
ArubaOS Version 8.5.0.5 (build 73491 / label #73491)
Built by yfbuild@hp-hpa-build05 on 2019-12-06 at 22:56:30 UTC (gcc version 4.9.4)
(12-04:11):Installing packages
Success: Package default_alngrp_pkg installed successfully
Success: Package default_uca_pkg installed successfully
Success: Package default_uaa_pkg installed successfully
Success: Package default_ara_cn_pkg installed successfully
Success: Package default_web_cc_pkg installed successfully
Success: Package default_ahapi_helper_pkg installed successfully
Success: Package default_alnmatch_pkg installed successfully
Success: Package default_appRF_pkg installed successfully
(12-04:12):Verify the bootloader [ OK ]
(12-04:12):rcS Done(17 sec)

(12-04:12):Starting OS services [ OK ]

(12-04:17):Initializing GSM [ DONE ]
(12-04:19):Initializing CCM [ DONE ]
(12-04:34):Initializing FPMPPs [ DONE ]
(12-04:52):Initializing CQM [ DONE ]
(12-04:57):Waiting for Controller IP [ DONE ]
(12-04:59):Initializing AAA [ DONE ]
(12-04:59):Initializing Controller management [ DONE ]

User: admin
Password:

User: admin
Password:

(CSM405-greeneshbasnet) [mgnode] show ip interface brief

Interface IP Address / IP Netmask Admin Protocol VRRP-IP
vlan 1 172.16.0.1 / 255.255.255.0 up up
loopback unassigned / unassigned up up
mgmt unassigned / unassigned down down
(CSM405-greeneshbasnet) [mgnode] show controller-ip

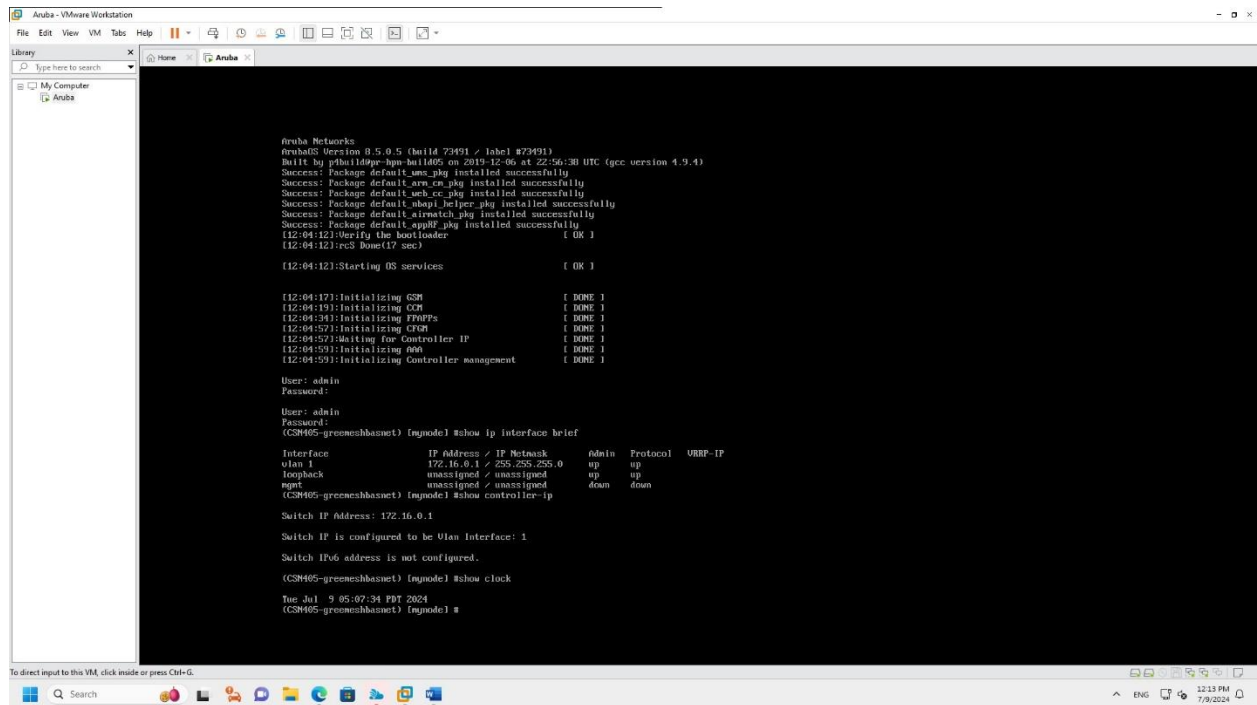
Switch IP Address: 172.16.0.1

Switch IP is configured to be Vlan Interface: 1

Switch IPv6 address is not configured.

(CSM405-greeneshbasnet) [mgnode] #
```

3. Show clock



The screenshot shows the Aruba VMWare Workstation interface. The main window displays the output of the 'show clock' command in a terminal window. The output shows the system time as Tue Jul 9 05:07:34 PDT 2024. The terminal also shows the output of 'show ip interface brief' and 'show controller-ip' commands.

```
Aruba Networks
ArubaOS Version 8.5.0.5 (build 73491 / label #73491)
Built by pfbuilder@aruba on 2019-12-06 at 22:56:30 UTC (gcc version 4.9.4)
Success: Package default_smc_pkg installed successfully
Success: Package default_arp_en_pkg installed successfully
Success: Package default_web_en_pkg installed successfully
Success: Package default_802.1x_helper_pkg installed successfully
Success: Package default_802.1x_match_pkg installed successfully
Success: Package default_appfw_pkg installed successfully
12:04:12:Verify the bootloaders ( OK )
12:04:12:rcs done(17 sec)

12:04:12:Starting OS services ( OK )

12:04:17:Initializing GSM ( DONE )
12:04:19:Initializing CCM ( DONE )
12:04:34:Initializing FPMPP ( DONE )
12:04:57:Initializing CFM ( DONE )
12:04:57:Waiting for Controller IP ( DONE )
12:04:59:Initializing aaaa ( DONE )
12:04:59:Initializing Controller management ( DONE )

User: admin
Password:

User: admin
Password:
(CSM405-greenmeshbaset) (mynode) #show ip interface brief

Interface      IP Address / IP Netmask      Admin  Protocol  VRRP-IP
-----
vlan 1         172.16.0.1 / 255.255.255.0    up      up
loopback       unassigned / unassigned      up      up
mgmt           unassigned / unassigned      down    down
(CSM405-greenmeshbaset) (mynode) #show controller-ip

Switch IP Address: 172.16.0.1

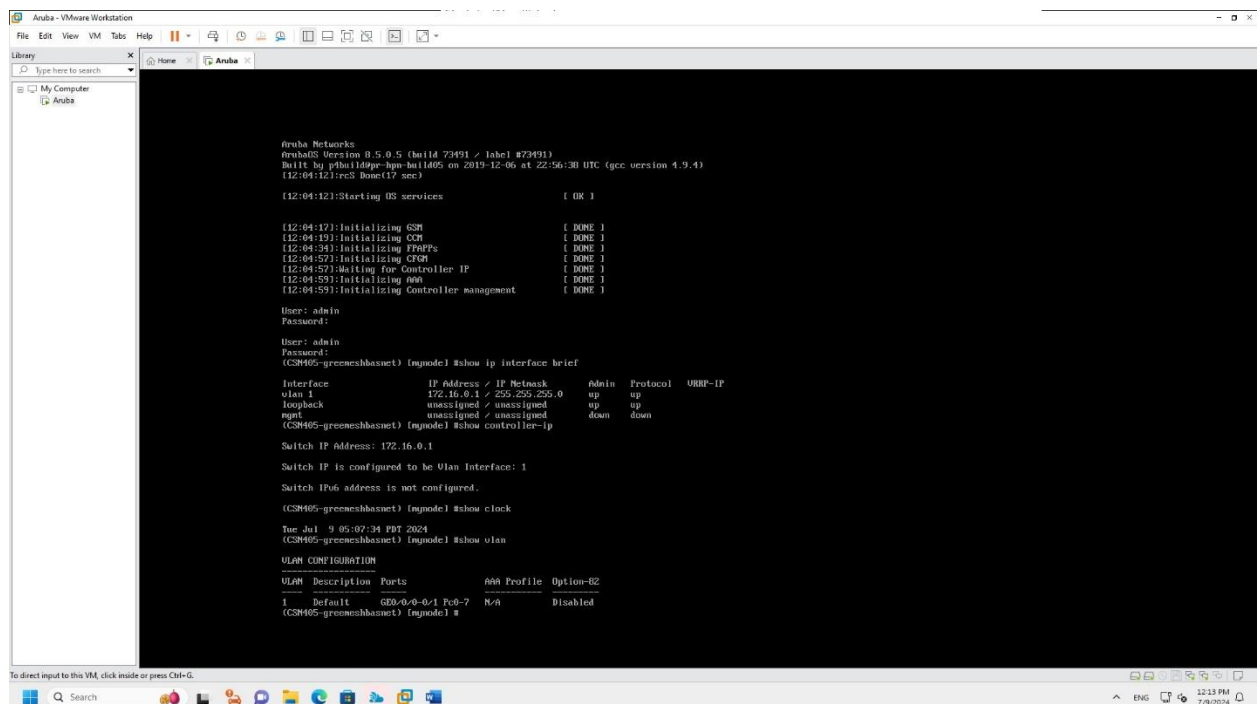
Switch IP is configured to be Vlan Interface: 1

Switch IPv6 address is not configured.

(CSM405-greenmeshbaset) (mynode) #show clock

Tue Jul 9 05:07:34 PDT 2024
(CSM405-greenmeshbaset) (mynode) #
```

4. Show vlan



The screenshot shows the Aruba VMWare Workstation interface. The main window displays the output of the 'show vlan' command in a terminal window. The output shows the VLAN configuration for the switch, including the VLAN ID, description, ports, and other details.

```
Aruba Networks
ArubaOS Version 8.5.0.5 (build 73491 / label #73491)
Built by pfbuilder@aruba on 2019-12-06 at 22:56:30 UTC (gcc version 4.9.4)
12:04:12:rcs done(17 sec)

12:04:12:Starting OS services ( OK )

12:04:17:Initializing GSM ( DONE )
12:04:19:Initializing CCM ( DONE )
12:04:34:Initializing FPMPP ( DONE )
12:04:57:Initializing CFM ( DONE )
12:04:57:Waiting for Controller IP ( DONE )
12:04:59:Initializing aaaa ( DONE )
12:04:59:Initializing Controller management ( DONE )

User: admin
Password:

User: admin
Password:
(CSM405-greenmeshbaset) (mynode) #show ip interface brief

Interface      IP Address / IP Netmask      Admin  Protocol  VRRP-IP
-----
vlan 1         172.16.0.1 / 255.255.255.0    up      up
loopback       unassigned / unassigned      up      up
mgmt           unassigned / unassigned      down    down
(CSM405-greenmeshbaset) (mynode) #show controller-ip

Switch IP Address: 172.16.0.1

Switch IP is configured to be Vlan Interface: 1

Switch IPv6 address is not configured.

(CSM405-greenmeshbaset) (mynode) #show clock

Tue Jul 9 05:07:34 PDT 2024
(CSM405-greenmeshbaset) (mynode) #show vlan

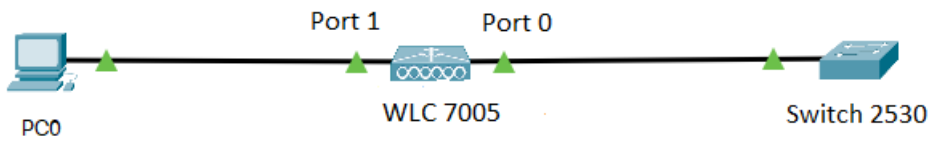
VLAN CONFIGURATION

VLAN  Description  Ports  AAA Profile  Option-82
----  -
1      Default  GE0/0-0-0/1 Fe0-7  N/A          Disabled
(CSM405-greenmeshbaset) (mynode) #
```

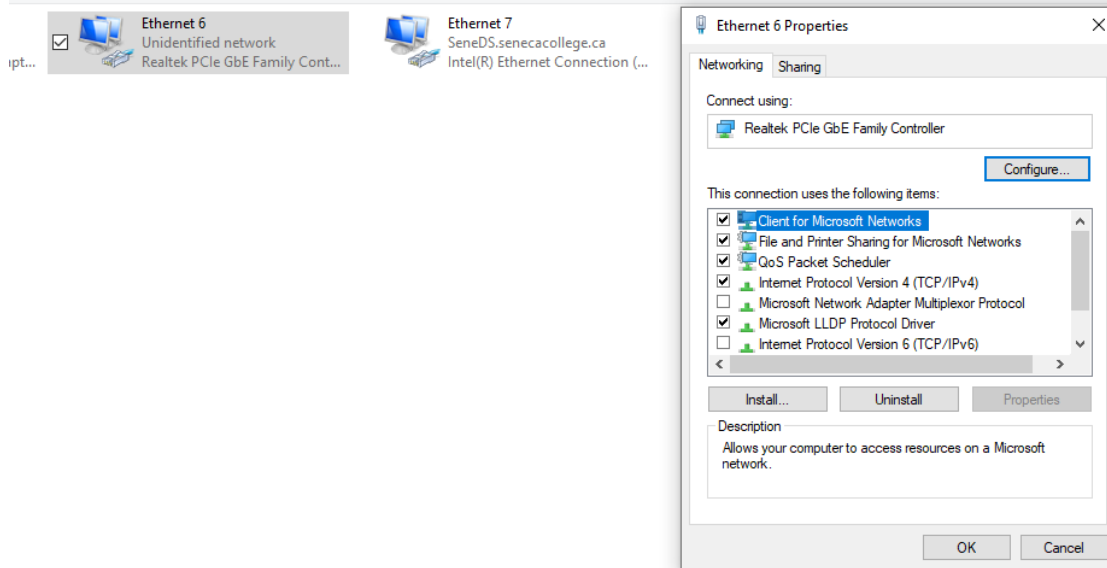
Part 2: Configuration using GUI

To configure the controller using GUI, follow the following steps:

1. Connect a straight through cable from NIC on PC using RJ45 connector to Port 1 on the controller.



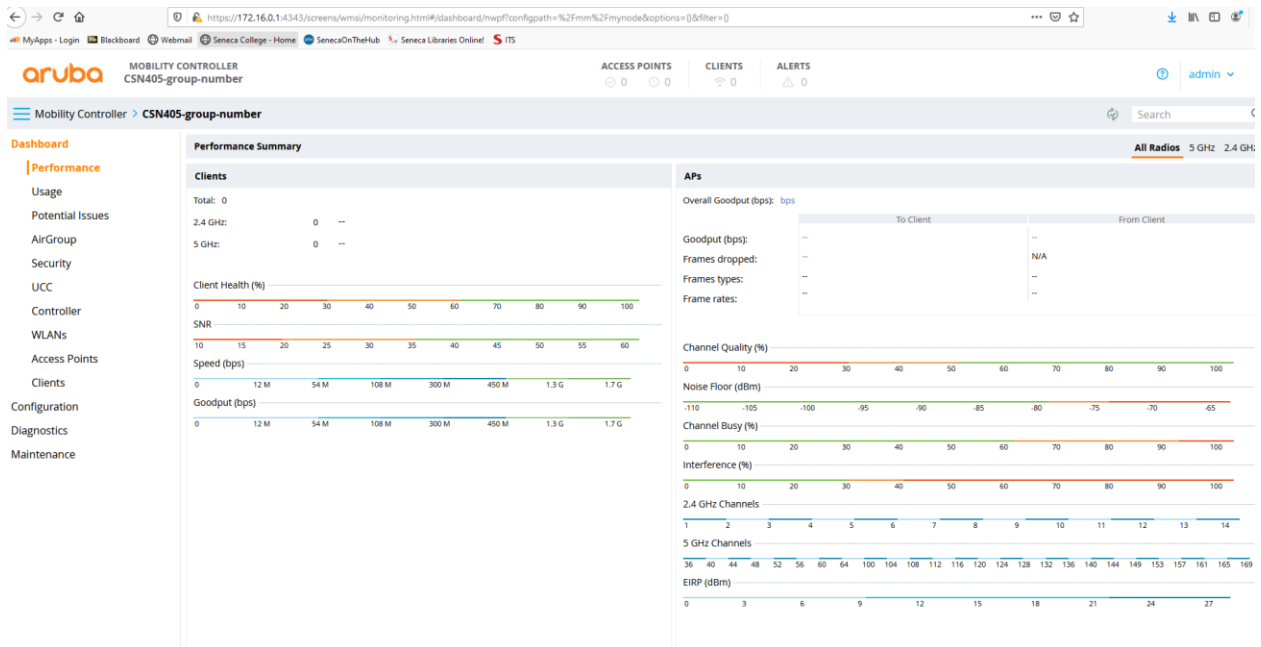
2. Make sure you are in the same subnet of the controller (assign an appropriate IP address, for example 172.16.0.6)



3. Make the network changes to connect to controller.
4. Open up the browser and access the controller using the controller IP.
5. Now, you will get the following page:



6. After logging-in, you will have prompted with the following window:



7. Click on Controller and answer the following questions:

During the lab, no controller was found.

1. MAC address start:
2. MAC address end:
3. System time:
4. Software:
5. Deployment:

8. Now, click on *Events* and provide screenshot for the result.

As the Aruba homepage does not appear to have a controller. I have therefore not captured the two screenshots that follow as instructed.

Aruba Mobility Master CSN405-greemeshbasnet

CONTROLLERS 0 ACCESS POINTS 0 CLIENTS 0 ALERTS 0

admin

Managed Network

Dashboard

Configuration

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces
- Controllers
- System
- Tasks
- Redundancy

Maintenance

ArubaMM-VA_8.5.0.5

Aruba Mobility Master CSN405-greemeshbasnet

CONTROLLERS 0 ACCESS POINTS 0 CLIENTS 0 ALERTS 0

admin

Managed Network

Dashboard

- Overview
- Infrastructure
- Security
- Services

Configuration

Maintenance

ArubaMM-VA_8.5.0.5

No controller found in selected group.

There are no controllers for the last part.

Answer the following questions:

(Important Note: Cite your reference(s) used to answer the questions.)

1. List the deployment modes of Aruba WLC Controller. Explain each one briefly.

There are two types of deployment mode of Aruba WLC Controller. They are: Master or Local.

Master Controllers: Features like WIPS, initial AP configurations, user roles, authentication-related configurations, and so forth are included in the master controller.

Local Controller: The local controller controls mobility features, QOS, adaptive radio management, shutdown of AP tunnels, and processing and forwarding of user traffic, including authentication.

2. How much power is required to run the Aruba Controller 7005? What is the IEEE 802.3 PoE standard is used for that purpose? Discuss briefly.

According to the information on the internet, the Aruba Controller 7005 needs DC 12V power to operate. The reason for using the IEEE 802.3 power over Ethernet standard is that it makes installation easier by enabling data and power access over a single Ethernet cable. Up to 15.4 watts of power per port are available with PoE (802.3af), which can be compacted to the lowest power needed by the 7005 controller.

3. Can the Access Point be connected to the Controller port directly? Explain why.

Since Aruba Controller's ports are made to offer network connectivity and management features for linked Aps, informal communications and centralised control of wireless networks are made possible, the Access Point and Controller can be connected directly to one another.

4. What is the maximum throughput can be achieved with Aruba 7005 Controller? Explain.

With the Aruba 7005 Controller, a throughput of up to 4 Gbps is possible. A wired bridging throughout of 4 Gbps. A firewall throughout of 4 Gbps and an encrypted throughout of 1.2 Gbps are all pushed by the 7005 Controller.

5. What are the differences between Mobility Master (MM) and Mobility Controller (MC)? Discuss.

Between Mobility Master and Mobility Controller, the following distinctions exist:

As a result of growing demand for mobile and Internet of things devices, mobility master enable customers to take advantage of advanced features that call for central coordination for networks to scale. Mobility controllers function as local controllers, allowing clients to roam within the mobility domain.

As a centralized management system, mobility master normally resides in a data center, while mobility master normally resides in a data center, while mobility controller is typically installed at the edge to handle access directly.

The Mobility master offers sophisticated functionality and scalability for big and complicated networks, while the Mobility Controller concentrates on local administration.